



UNDERSTANDING DATA VISUALIZATION



DATA VISUALIZATION OVERVIEW



Definition

Data visualization is the practice of presenting complex data and information visually using elements like charts, graphs, maps, and dashboards to make patterns, trends, and outliers easily understandable, enabling quicker insights and better decision-making for both technical and non-technical audiences.



Importance

- Simplifies Complex Data:** Converts large data into easy-to-understand charts and graphs
- Reveals Patterns and Trends:** Highlights trends and relationships clearly
- Saves Time:** Allows quick understanding of important information
- Improves Communication:** Makes data easier to explain to others
- Tells a Clear Story:** Helps guide decisions with clear insights



TYPES OF GRAPH

1. Bar Chart

- Used to compare values between different categories
- Example: Sales of different products

2. Pie Chart

- Shows parts of a whole (percentages)
- Example: Market share of companies

3. Line Graph

- Shows changes over time
- Example: Temperature changes during a week

4. Histogram

- Displays data distribution
- Example: Exam score ranges

5. Scatter Plot

- Shows relationship between two variables
- Example: Study time vs. exam score

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TOOLS USED FOR DATA VISUALIZATION



Microsoft Excel – Simple charts and graphs



Google Sheets – Online chart creation



Tableau – Advanced interactive dashboards



Power BI – Business data visualization



Python – Programming-based charts

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Best practices in Data Visualization.



•Know Your Audience

Build visualizations that meet the needs of real decision-makers.



•Choose the Right Visualization

Match the chart type to the data and purpose.



•Embrace Simplicity

Focus only on important information and remove unnecessary details.



•Use Color Intentionally

Use colors to highlight meaning, not to distract.



•Tell a Compelling Story

Add context and guide users through the data logically.



•Leverage Modern Tools

Use tools like Power BI, Tableau, and Looker for advanced visuals.



•Test, Iterate, and Improve

Collect feedback and continuously refine visualizations.

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THANK YOU

